Welcome to MATH 243

Structure: worksheet posted in advance of discussion. Attendance taken start of class via sheet passed around. Logistics remarks covered, students can ask general questions. 15-20 min at start for students to catch up on worksheet. After that, review begins.

Go over problems students request, then time permitted, select 1-2 problems illustrative of tricky concepts to cover.

Attendance: Taken every week after 1st week, each discussion counts equally.

Coming up: Class website to show materials, communication info, schedule, and anything else relevant to discussions, will be released by next week. Anonymous feedback survey happening in a few weeks. Besides the survey, feel free to share feedback at any time.

Worksheet: Available on Canvas this week. After that, available on class site, along with solutions posted after the last section that week ends.

3, ans 8, 4, ans ___,

3: Find equation of tangent at $x=1$ for $y=\sqrt{4+3f(x)}$ if $f(1)=7$ & $f'(1)=4$.

We need to find slope of line & point on line.

When $x=1$, we have $y = \sqrt{4+3f(1)} = \sqrt{4+3\cdot 7} = \sqrt{25} = 5$, so $(1,5)$ works.

$$\frac{dy}{dx} = \frac{3f'(x)}{2\sqrt{4+3f(x)}}$$ by using chain rule on $\sqrt{\cdots}$.

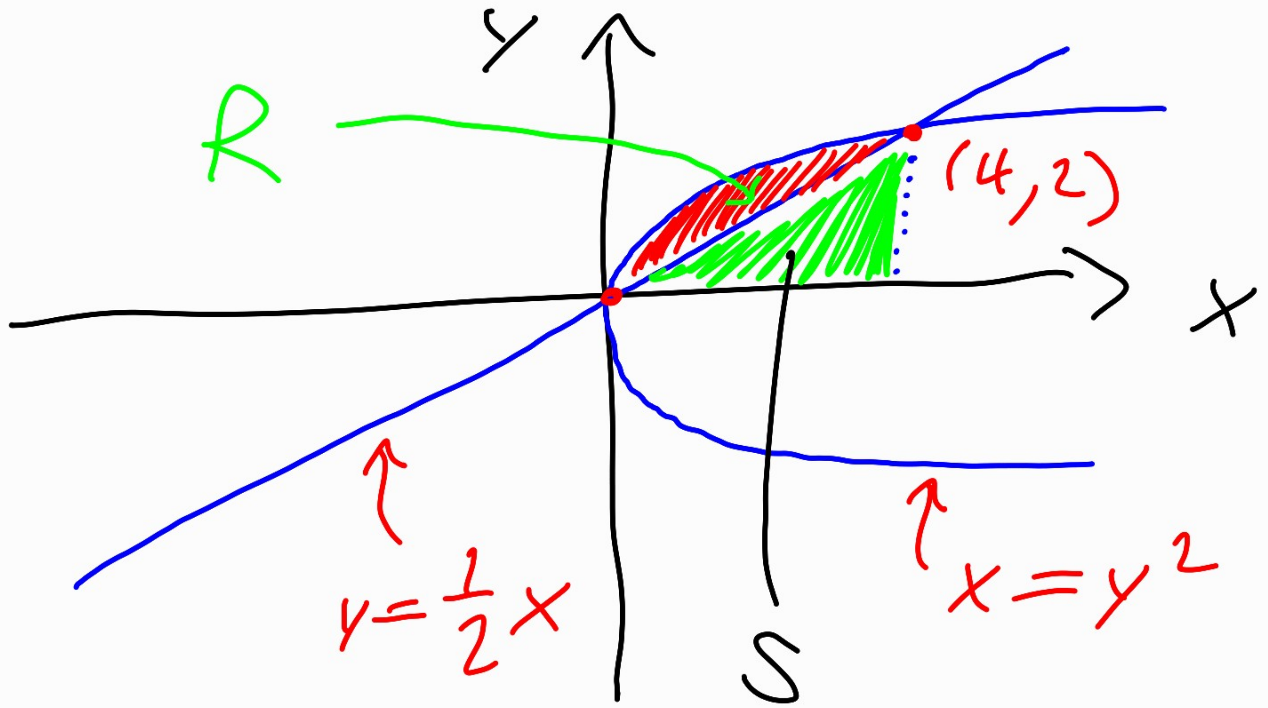
$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{3f'(1)}{2\sqrt{4+3f(1)}} = \frac{12}{2\cdot 5}$$

$= \frac{6}{5}$, so slope $= \frac{6}{5}$.

So line is $\boxed{y-5 = \frac{6}{5}(x-1)}$.

8: R bounded by $x=y^2$, $y=\frac{1}{2}x$

(a)



(b) Let S be triangle below R. Then $\text{area}(R) = \text{area}(S \cup R) - \text{area}(S)$.

$S \cup R$ is bounded by $y=\sqrt{x}$, S bounded by $y=\frac{1}{2}x$, and both regions go from $x=0$ to $x=4$. So the area is $\int_0^4 \left(\sqrt{x} - \frac{1}{2}x\right) dx$

$$= \left(\frac{2}{3}x^{3/2} - \frac{1}{4}x^2\right)\Big|_0^4 =$$

$$\frac{2}{3}\cdot 8 - \frac{1}{4}\cdot 16 = \frac{16}{3} - 4 = \frac{4}{3}$$

5b: $\int (2x^4+1)\ln x \, dx =$

$u = \ln x \qquad du = \frac{1}{x}$

$dv = (2x^4+1)\,dx \qquad v = \frac{2}{5}x^5 + x$

$$\int u\,dv = uv - \int v\,du$$

$$\left(\frac{2}{5}x^5 + x\right)\ln x - \int\left(\frac{2}{5}x^4 + 1\right) =$$

$$\boxed{\left(\frac{2}{5}x^5 + x\right)\ln x - \frac{2}{25}x^5 - x + C}$$

4: $f = x^3 - 3x^2 - 9x + 2,$

find all crit points &
find which are local min/max.

Recall: critical points are those where f' undefined or $f'=0$

$0 = f' = 3x^2 - 6x - 9 =$
$3(x^2 - 2x - 3) = 3(x-3)(x+1)$

$\Rightarrow \quad x = 3, -1.$

To check min/max, check f''.

$f'' = 6x - 6 = 6(x-1).$

$f''(3) = 12 > 0$, so $x=3$ is local min

$f''(-1) = -12 < 0$, so $x=-1$ is local max.

$(3, f(3))$ local min

$(-1, f(-1))$ is local max

1. $f' = -\sin(x^2+e^x)(2x+e^x)$

2. $\frac{dy}{dx} = -\frac{y\ln y}{x+ye^y}$

5a. $= e^{\tan x} + C$

b. $\left(\frac{2}{5}x^5+x\right)\ln x - \left(\frac{2}{25}x^5+x\right)+C$

6a. $\frac{1}{3}$

b. 1

7. $s(t) = -3\cos t + 2\sin t + 2t + 3$

9.